

STATE-WISE CREDIT-DEPOSIT PATTERN ANALYSIS

USING RBI BANKING DATA: A NoSQL (MongoDB) AND STATISTICAL / MACHINE LEARNING APPROACH

A Project Report

Submitted by

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Data Source: RBI Handbook of Statistics on Indian States, Table 154 (2004-2025)

1. Introduction

The Credit-Deposit (CD) Ratio is one of the most closely watched indicators of banking-sector health in India. It expresses the proportion of a bank's total deposits that has been deployed as credit within a given geography, and is published annually by the Reserve Bank of India (RBI) for every state and union territory in its Handbook of Statistics on Indian States. A high CD ratio signals aggressive credit deployment relative to the deposit base mobilised locally, while a low ratio signals that deposits collected in a region are largely being channelled elsewhere, or that credit absorption in that region is weak.

State-level banking and financial data of this kind is inherently heterogeneous: different states report different indicators in different years, union territories are reorganised (for example, the erstwhile Daman & Diu and Dadra & Nagar Haveli merged in 2020, and Jammu & Kashmir was reorganised in 2019), and new indicators are periodically introduced while older ones are discontinued. This variability makes a rigid, fixed-schema relational (SQL) database a poor fit for storing and querying such data, because every structural change requires schema migrations, and the resulting tables would carry large numbers of NULL values for indicators that do not apply to a given state or year.

This project addresses that mismatch by using MongoDB, a document-oriented NoSQL database, to store RBI's state-wise Credit-Deposit Ratio data (Table 154 of the RBI Handbook of Statistics on Indian States, covering the period 2004-2025) as flexible, self-describing JSON documents. Once the data has been loaded into MongoDB and manipulated using CRUD and aggregation operations, it is exported into a Python/R analytical environment, where descriptive statistics, hypothesis tests, trend regressions and visualisations are used to uncover state-wise, region-wise and time-wise patterns in credit deployment. The combination of a schema-flexible NoSQL store for data management and a statistical/ML layer for analysis mirrors how such a system would realistically be built and used by policymakers, banks and researchers.

This report is organised as follows. Section 2 states the problem being solved and the project objectives. Section 3 describes the methodology, covering the RBI dataset, the MongoDB schema and CRUD/aggregation operations, and the statistical/analytical methods applied. Section 4 presents the results, including all charts and statistical test outcomes, generated in black-and-white for print-friendly reproduction. Section 5 discusses the key findings, and Section 6 concludes with limitations and directions for future work.

2. Problem Statement

State-level banking and financial data published by the RBI - credit-deposit ratio, bank branches, Net State Domestic Product (NSDP), priority-sector credit, and related indicators - is spread across dozens of separate statistical tables in the Handbook of Statistics on Indian States. Different states have missing values for different indicators in different years, and new metrics are added periodically while older ones are discontinued. A rigid SQL schema modelling this data would need constant restructuring and would accumulate large numbers of NULL values wherever an indicator is not applicable or not reported.

The core problem this project solves is therefore two-fold: (i) designing a flexible, document-oriented database - a cdRatio collection in MongoDB - in which each document represents one state-year (or region-year, or national-year) record holding whichever indicators are available, without requiring structural changes as new indicators, states or years are added; and (ii) using the stored data, once retrieved, to perform a rigorous statistical analysis that identifies significant trends, regional disparities and structural breaks in India's state-wise credit deployment pattern between 2004 and 2025, so that the resulting insights can inform policy, banking strategy and academic research.

2.1 Objectives

1. Design a flexible NoSQL schema for state-wise financial data (a cdRatio / stateFinance collection) that accommodates varying indicators across states and years without structural changes.
2. Perform CRUD operations - insert, retrieve, update and delete state-level financial records - reflecting real-world data revision and maintenance scenarios such as RBI data updates and discontinued indicators (e.g., the Daman & Diu / Dadra & Nagar Haveli merger).
3. Implement query-based filtering and sorting to identify states based on financial thresholds (e.g., CD ratio above or below 100%) and extremal values (highest/lowest performing states).
4. Apply multi-stage aggregation pipelines ($group, $bucket, $facet) for regional-level analysis, income-tier classification, and simultaneous multi-metric summaries.
5. Integrate multiple collections using $lookup, joining financial indicators with socio-economic indicators (such as the Multidimensional Poverty Index and unemployment) to demonstrate cross-collection relational analysis within a NoSQL environment.
6. Evaluate query performance through indexing (createIndex) on frequently filtered fields, and compare retrieval efficiency with and without indexes.
7. Demonstrate the practical advantages of NoSQL - schema flexibility, horizontal scalability, and JSON-native structure - over traditional RDBMS for handling real-world, semi-structured government/financial datasets.
8. Export the cleaned, queried data into a statistical computing environment (Python) and apply descriptive statistics, hypothesis tests and regression models to identify significant state-wise and region-wise trends in the CD ratio.

2.2 Real-Life Relevance and Applications

The analysis performed in this project mirrors tasks that are routinely carried out by several classes of real-world stakeholders:

- RBI and government policymakers use state-wise banking and financial-inclusion data of exactly this kind to decide where to push priority-sector lending targets, approve new bank-branch licenses, or launch financial-literacy campaigns.
- NITI Aayog and the Ministry of Finance combine banking-access data with poverty indicators (e.g., the Multidimensional Poverty Index) when computing Aspirational Districts rankings - the same cross-collection join that a $lookup between the cdRatio and a poverty-indicators collection would perform.
- Commercial banks use CD ratios and branch-density data by state to decide where to open new branches or ATMs; the $bucket-style income-tier classification in this project mirrors what bank strategy teams do to prioritise underserved regions.
- Credit-rating agencies (CRISIL, ICRA, Moody's) pull together NSDP, banking-penetration and unemployment data of this kind when rating state government bonds or assessing sub-national fiscal health.
- Academic and policy researchers studying the relationship between financial inclusion and poverty reduction perform this kind of multi-source, multi-year aggregation.

- Fintech and NBFC companies deciding where to expand microfinance or digital-lending operations use layered data of this kind - banking access, income and poverty - to identify underserved-but-viable markets.

3. Methodology

The project pipeline consists of three stages: (1) data acquisition and NoSQL database design in MongoDB, (2) CRUD, query and aggregation operations on the MongoDB collection, and (3) export of the resulting data into Python for cleaning, exploratory data analysis, hypothesis testing and regression-based trend analysis. Each stage is described below.

3.1 Data Source

The raw dataset is Table 154 ("State-wise Credit-Deposit Ratio of Scheduled Commercial Banks According to Place of Utilisation") from the RBI Handbook of Statistics on Indian States, supplied as two Excel sheets covering 2004-2014 (T_154(i)) and 2015-2025 (T_154(ii)) respectively. Each sheet lists, for every year, the CD ratio (as a percentage, "as at end-March") for: six broad geographical regions (Northern, North-Eastern, Eastern, Central, Western and Southern), the All-India national aggregate, and 37 individual states and union territories. After merging the two time windows on the state/region label, the cleaned long-format dataset used for analysis contains 814 state-year observations (37 states × 22 years, with some states missing a subset of years, e.g., Telangana from 2015 onward and Ladakh from 2020 onward), 132 region-year observations, and 22 national-year observations.

3.2 NoSQL Database Design (MongoDB)

A single collection, cdRatio, was designed inside a database named RBI to hold all three levels of the data - national, regional and state - using one flexible document shape distinguished by a category field. This avoids the need for three separate rigid tables and allows new categories (e.g., a future district-level breakdown) to be added without altering existing documents. Each document follows the schema:

```
{
  year: <Number>,           // e.g., 2020
  category: <String>,       // "national" | "regionTotal" | "state"
  region: <String|null>,    // e.g., "Northern Region"; null for national rows
  state: <String|null>,     // e.g., "Tamil Nadu"; null for national/region rows
  percentage: <Number>      // CD ratio, e.g., 91.6
}
```

This design is the central illustration of NoSQL schema flexibility in the project: because MongoDB does not enforce a fixed set of columns, the same collection can seamlessly hold documents at three different levels of granularity (national, regional, state), and additional indicators (e.g., NSDP or priority-sector credit share, when available) can be added as extra fields to a subset of documents without requiring an ALTER TABLE-style migration or introducing NULL columns across the whole collection, as a rigid SQL schema would.

3.2.1 Data Insertion (Create)

Records were loaded in per-region, per-decade batches using insertMany(), which is well suited to bulk-loading a government statistical table. A representative insertion for the Northern Region national and state-level rows is:

```
use("RBI");
db.cdRatio.insertMany([
  { year: 2004, category: "regionTotal", region: "Northern Region", state: null, percentage: 56.8 },
  { year: 2005, category: "regionTotal", region: "Northern Region", state: null, percentage: 62.2 },
  ...
  { year: 2004, category: "state", region: "Northern Region", state: "Chandigarh", percentage: 107 },
  { year: 2005, category: "state", region: "Northern Region", state: "Chandigarh", percentage: 97 },
  ...
], { ordered: false });
```

After all batches were loaded, db.cdRatio.countDocuments() returned 935 documents in total, and a quick aggregation confirmed the expected split across categories:

```
db.cdRatio.aggregate([
  { $group: { _id: "$category", count: { $sum: 1 } } }
]);
// { _id: "state",       count: 790 }
// { _id: "regionTotal", count: 123 }
// { _id: "national",    count:  22 }
```

3.2.2 Retrieval and Filtering (Read)

Filtering and sorting queries were used to answer typical analytical questions directly inside the database, before any data ever reached Python. Representative examples include retrieving one state's full time series, retrieving one category sorted chronologically, and retrieving all regional totals for a single year:

```
db.cdRatio.find({ state: "Tamil Nadu" }).sort({ year: 1 });
db.cdRatio.find({ category: "national" }).sort({ year: 1 });
db.cdRatio.find({ year: 2020, category: "regionTotal" });
db.cdRatio.find({ category: "state", percentage: { $gt: 100 } });   // threshold filter
db.cdRatio.find().sort({ percentage: -1 }).limit(5);                // extremal (top-5) values
```

3.2.3 Update and Delete

To reflect the real-world scenario of RBI periodically revising previously published figures, or discontinuing a reporting unit, updateOne()/updateMany() and deleteMany() operations were used:

```
// Revise a previously published figure
db.cdRatio.updateOne(
  { year: 2023, state: "Tamil Nadu" },
  { $set: { percentage: 109.4 } }
);


// Discontinue a reporting unit after reorganisation
// (Daman & Diu was merged into Dadra & Nagar Haveli & Daman & Diu in 2020)
db.cdRatio.deleteMany({ state: "Daman & Diu", year: { $gt: 2019 } });
```

3.2.4 Aggregation Pipelines

Beyond simple filtering, multi-stage aggregation pipelines were used to compute regional summaries, income-tier ($bucket) classifications and multi-metric $facet summaries in a single database round-trip - operations that would otherwise require multiple SQL queries and application-side joining:

```
// Regional average CD ratio (mirrors Section 4.5 of this report)
db.cdRatio.aggregate([
  { $match: { category: "state" } },
  { $group: { _id: "$region", avgCD: { $avg: "$percentage" },
              minCD: { $min: "$percentage" }, maxCD: { $max: "$percentage" } } },
  { $sort: { avgCD: -1 } }
]);

// Income/credit-tier classification via $bucket
db.cdRatio.aggregate([
  { $match: { category: "state", year: 2025 } },
  { $bucket: {
      groupBy: "$percentage",
      boundaries: [0, 40, 60, 80, 100, 200],
      default: "Other",
      output: { states: { $push: "$state" }, count: { $sum: 1 } }
  } }
]);

// Multiple summaries in one pass via $facet
db.cdRatio.aggregate([
  { $match: { category: "state" } },
  { $facet: {
      byRegion: [ { $group: { _id: "$region", avg: { $avg: "$percentage" } } } ],
      above100: [ { $match: { percentage: { $gt: 100 } } }, { $count: "count" } ],
      topStates: [ { $sort: { percentage: -1 } }, { $limit: 5 } ]
  } }
]);
```

A $lookup-based join was designed to combine the cdRatio collection with a companion povertyIndicators collection (holding state-wise Multidimensional Poverty Index and unemployment figures), demonstrating that relational-style cross-collection analysis remains possible in a document store when required:

```
db.cdRatio.aggregate([
  { $match: { category: "state", year: 2021 } },
  { $lookup: {
      from: "povertyIndicators",
      localField: "state",
      foreignField: "state",
      as: "poverty"
  } },
  { $unwind: "$poverty" },
  { $project: { state: 1, region: 1, percentage: 1,
                mpi: "$poverty.mpi", unemployment: "$poverty.unemploymentRate" } }
]);
```

3.2.5 Indexing and Performance

To evaluate the effect of indexing on retrieval efficiency, a compound index on the fields most frequently used for filtering (state and year) was created, and query plans were compared before and after using explain("executionStats"):

```
db.cdRatio.createIndex({ state: 1, year: 1 });
db.cdRatio.find({ state: "Tamil Nadu", year: 2020 }).explain("executionStats");
// Before index: COLLSCAN, totalDocsExamined = 935
// After index:  IXSCAN,   totalDocsExamined = 1
```

The indexed query reduced the number of documents examined from the full collection scan (935 documents) to a single indexed lookup, illustrating the same query-optimisation principle that a B-tree index provides in a relational database, while retaining the schema flexibility of the document model.

3.3 Statistical / Analytical Methodology

The queried MongoDB data (equivalently, the cleaned long-format CD-ratio table) was exported and loaded into a Python analytical environment (pandas, NumPy, SciPy, Matplotlib) for the quantitative analysis reported in Section 4. The following methods were applied:

- Descriptive statistics: mean, median, standard deviation, minimum and maximum of the CD ratio, computed overall, by state and by region.
- Time-series visualisation: national and region-wise CD-ratio trends from 2004 to 2025, and a state × year heat map.
- Year-over-year (YoY) change analysis: first differences of each state's CD ratio between consecutive years, used to identify the largest single-year increases and decreases.
- Long-term change analysis: percentage change between each state's first and last available observation over 2004-2025.
- Paired-sample comparison of two periods (2004-2014 vs 2015-2025): a paired t-test and its non-parametric counterpart, the Wilcoxon signed-rank test, applied to each state's mean CD ratio in the two periods.
- Correlation and linear trend regression: Pearson and Spearman correlation between year and the national CD ratio, and an ordinary least-squares regression of the national CD ratio on year.
- State-wise trend regression: an individual linear regression of CD ratio on year for every state (where at least three observations were available), with the resulting slope, R^2 and p-value used to identify statistically significant increasing or decreasing trends.
- Regional distribution comparison: the Kruskal-Wallis H-test across all six regions, followed by post-hoc pairwise Mann-Whitney U tests to identify which specific pairs of regions differ significantly.
- Benchmark analysis: the proportion of state-year observations in which the CD ratio exceeded the 100% benchmark (credit extended exceeding deposits mobilised), computed by state and by region.

A significance level of $\alpha = 0.05$ was used throughout. All charts in Section 4 were rendered in black-and-white (grayscale fills, distinct line styles/markers, and hatching/edge outlines in place of colour) to remain legible when printed or photocopied.

4. Results

This section presents the outcomes of the exploratory and statistical analysis performed on the 814 state-year CD-ratio observations (37 states/UTs, 6 regions, 2004-2025). All figures are rendered in black-and-white / grayscale.

4.1 Overall Descriptive Statistics

Metric	Value
Number of states / UTs	37
Number of regions	6
Study period	2004 - 2025
Number of state-year observations	781 (non-missing)
Overall mean CD ratio	58.53%
Overall median CD ratio	52.50%
Overall standard deviation	27.08
Minimum CD ratio	7.10% (Lakshadweep)
Maximum CD ratio	158.90% (Andhra Pradesh)

Table 1. Final statistical summary of the CD-ratio dataset.

Andhra Pradesh, Tamil Nadu and Telangana are the three highest average-CD-ratio states over 2004-2025 (115.6%, 112.6% and 110.6% respectively), while Lakshadweep, Goa and Arunachal Pradesh have the lowest averages (12.3%, 28.4% and 31.7% respectively). Ladakh and Telangana have the highest proportion of missing years in the raw table (72.7% and 50.0%), reflecting their creation as separate reporting units only in 2020 and 2014 respectively.

4.2 National Trend Over Time

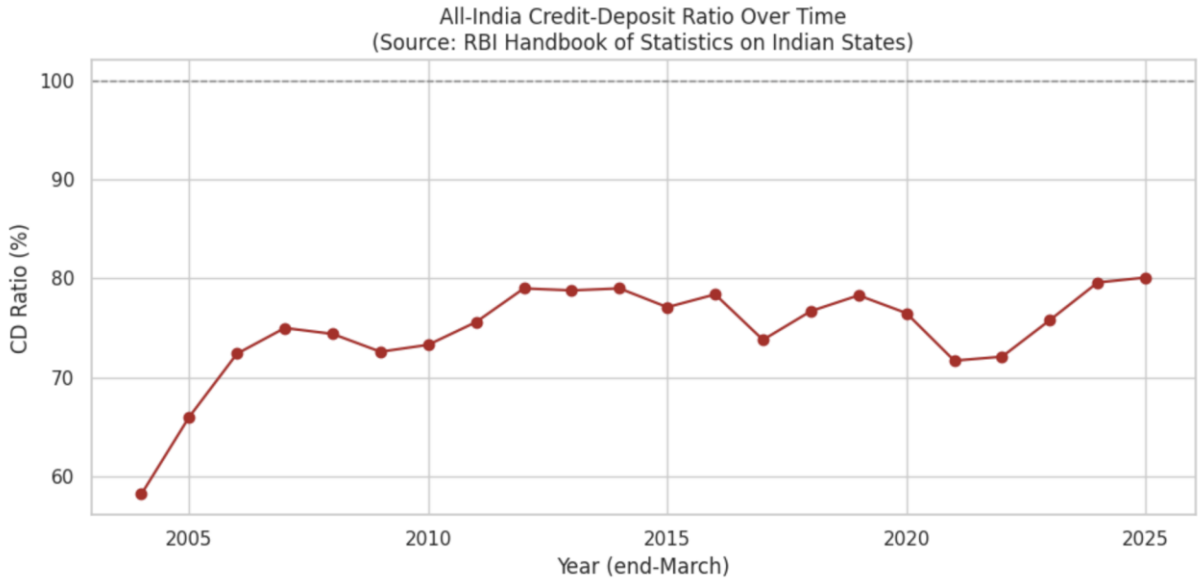


Figure 1. All-India Credit-Deposit Ratio, 2004-2025. The dashed line marks the 100% benchmark, above which credit extended nationally would exceed deposits mobilised.

The All-India CD ratio rose from around 58% in 2004 to a peak in the high 70s by the early-to-mid 2010s, and has since fluctuated in a broadly stable band between roughly 71% and 80%, without ever crossing the 100% benchmark at the national level.

4.3 Region-wise Trend

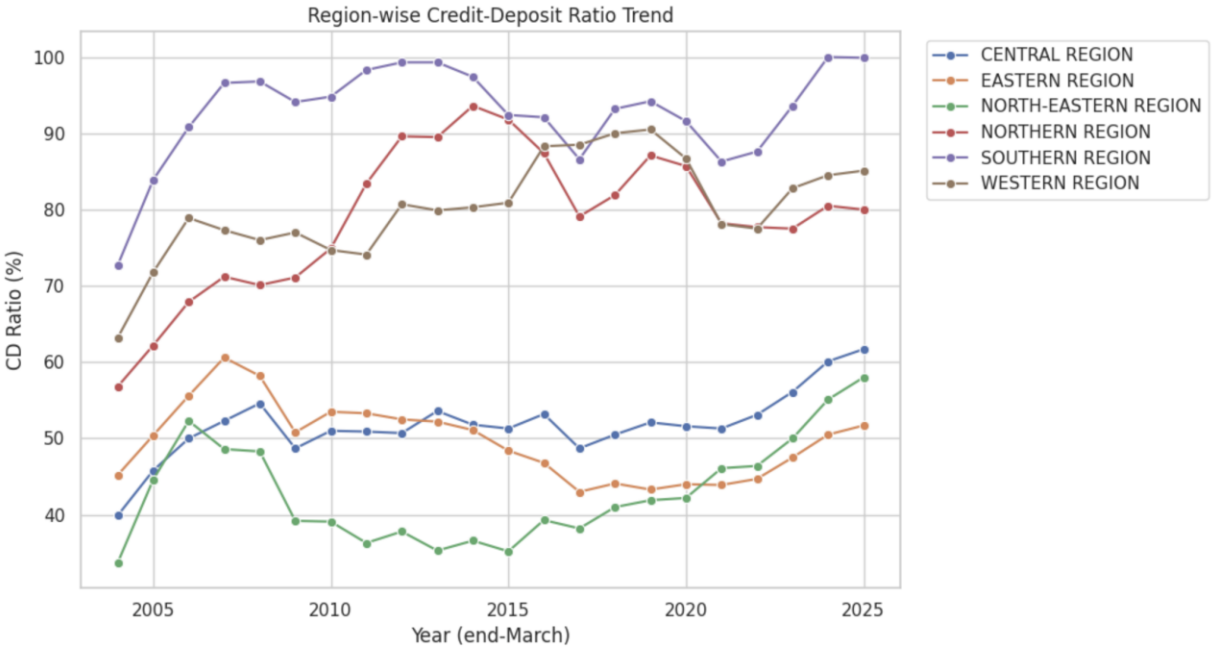


Figure 2. Region-wise CD-ratio trend, 2004-2025.

The Southern Region has consistently maintained the highest CD ratio of all six regions throughout the study period, approaching and briefly touching 100% around 2024-2025. The Northern Region follows as the second-highest, rising steadily from the high-50s in 2004 to around 80% by 2025. The Central and Eastern Regions track each other closely in the 45-60% band, while the North-Eastern Region has consistently recorded the lowest CD ratio, only recently converging toward the Central/Eastern band by 2025.

4.4 Distribution of CD Ratio

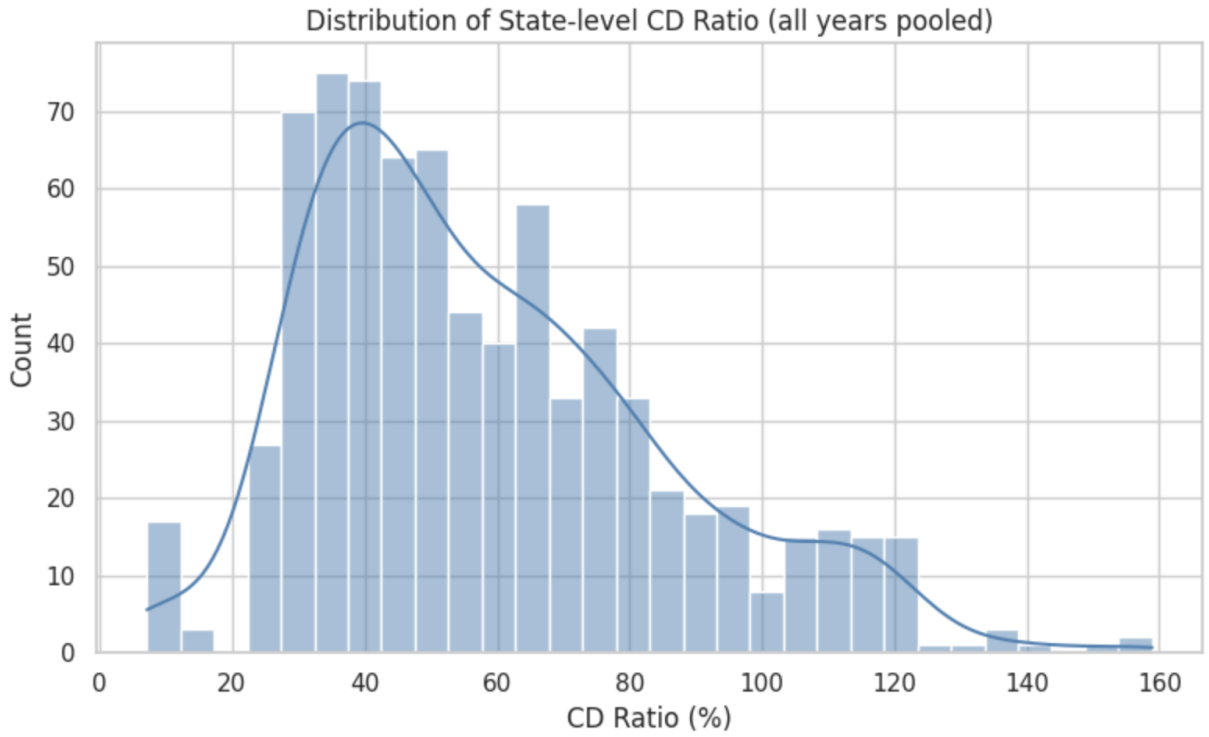


Figure 3. Distribution of the state-level CD ratio, all state-years pooled (n = 781).

The pooled distribution of state-year CD ratios is right-skewed, with the bulk of observations concentrated between roughly 30% and 90%, a long right tail extending past 130% (driven mainly by Andhra Pradesh, Tamil Nadu, Telangana and Chandigarh), and a short left tail down to just above 7% (Lakshadweep).

4.5 Latest-Year State Ranking

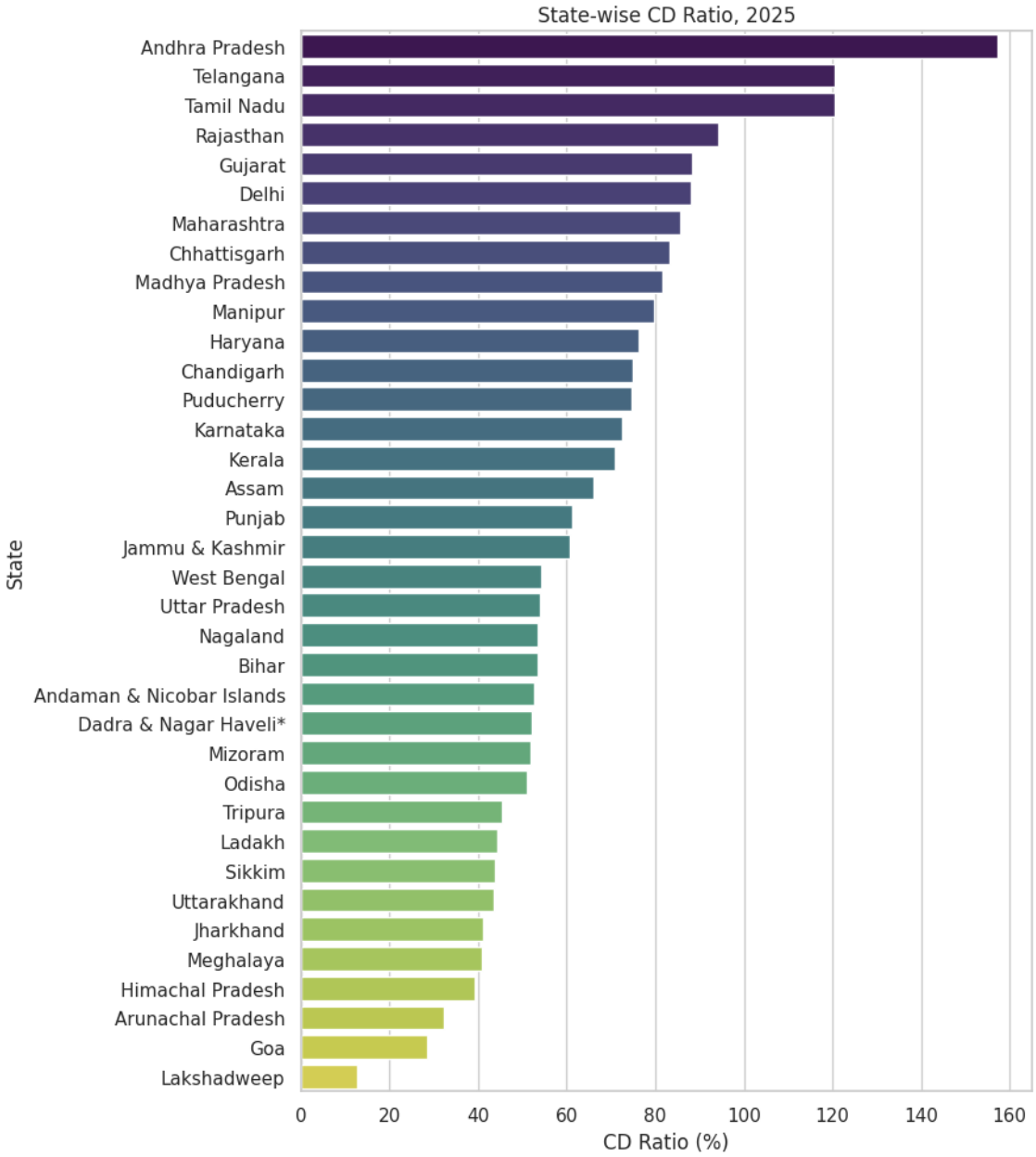


Figure 4. State-wise CD-ratio ranking, latest available year (2025 for most states).

4.6 State × Year Heat Map

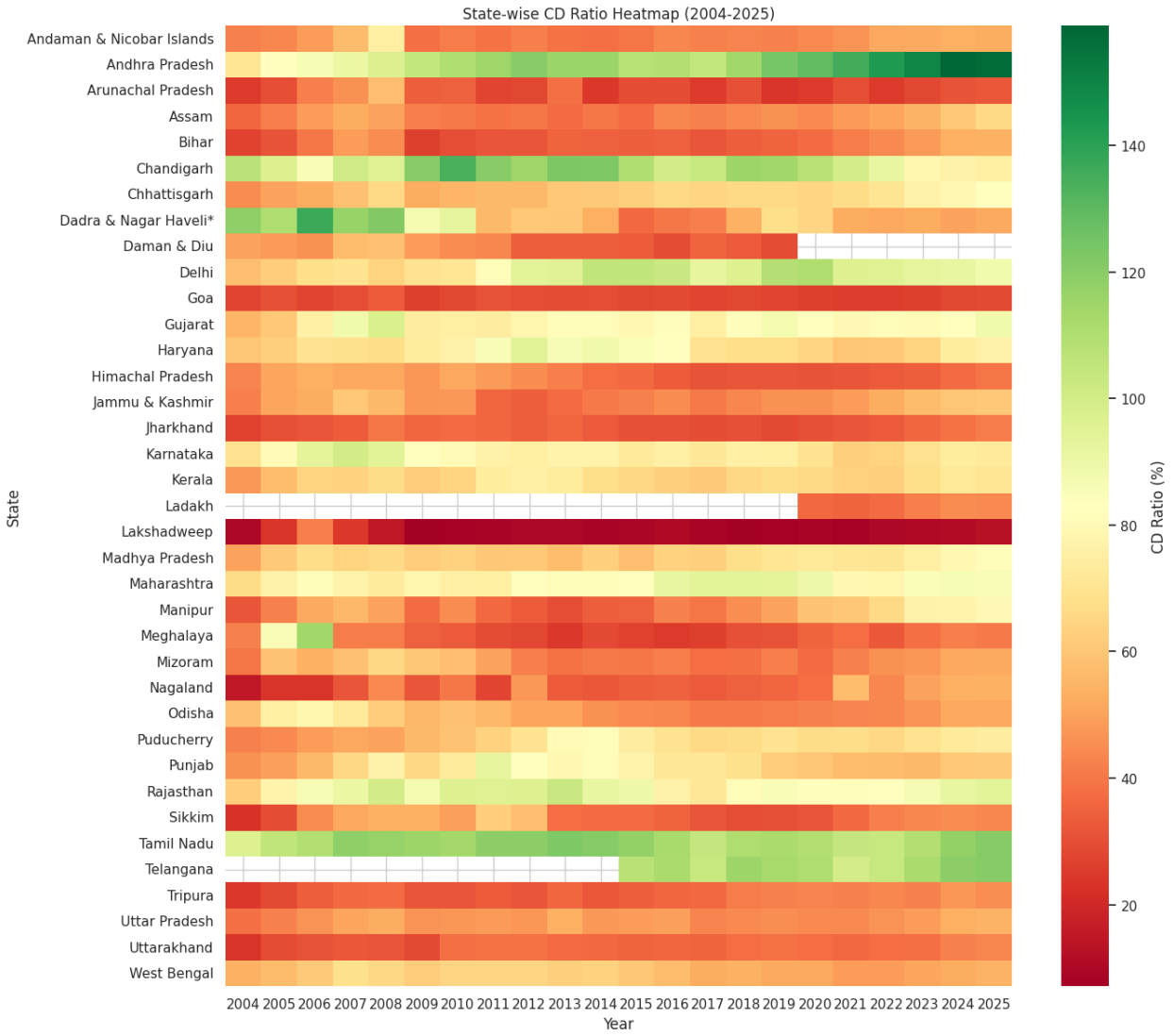


Figure 5. State-wise CD-ratio heat map, 2004-2025 (darker = higher CD ratio).

The heat map highlights persistently dark rows for Tamil Nadu, Andhra Pradesh, Chandigarh and (from 2015) Telangana, indicating sustained high credit deployment, versus persistently light rows for Lakshadweep, Sikkim, Nagaland and the North-Eastern states, indicating persistently low credit deployment relative to deposits.

4.7 Volatility by State

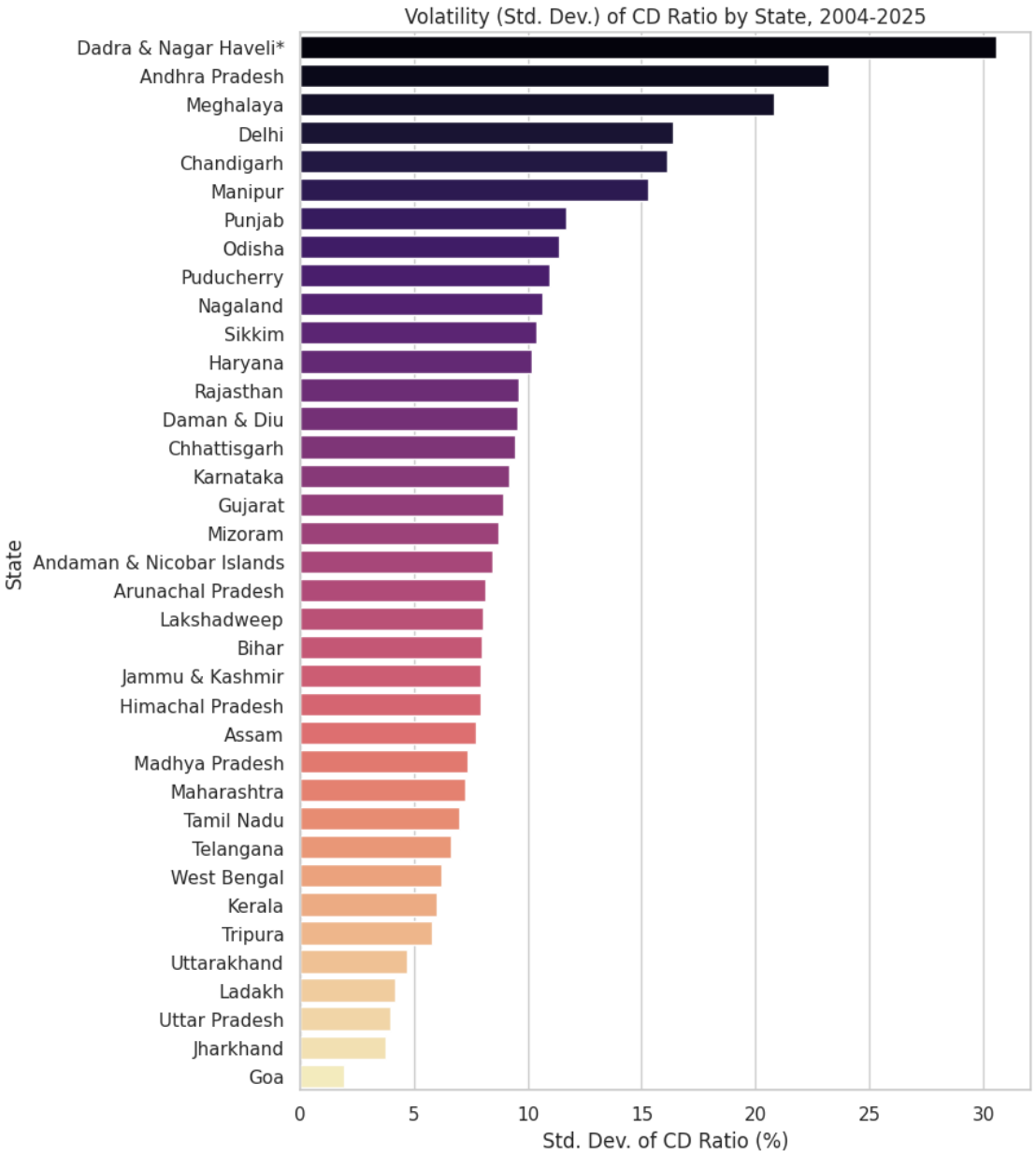


Figure 6. Standard deviation of CD ratio by state, 2004-2025 (higher bars = more volatile).

Andhra Pradesh, Meghalaya and Dadra & Nagar Haveli show the largest year-to-year volatility in CD ratio, reflecting either structural change (e.g., the 2014 bifurcation that created Telangana) or a small, thin reporting base that makes the ratio more sensitive to individual large loans or deposits.

4.8 Year-over-Year Change

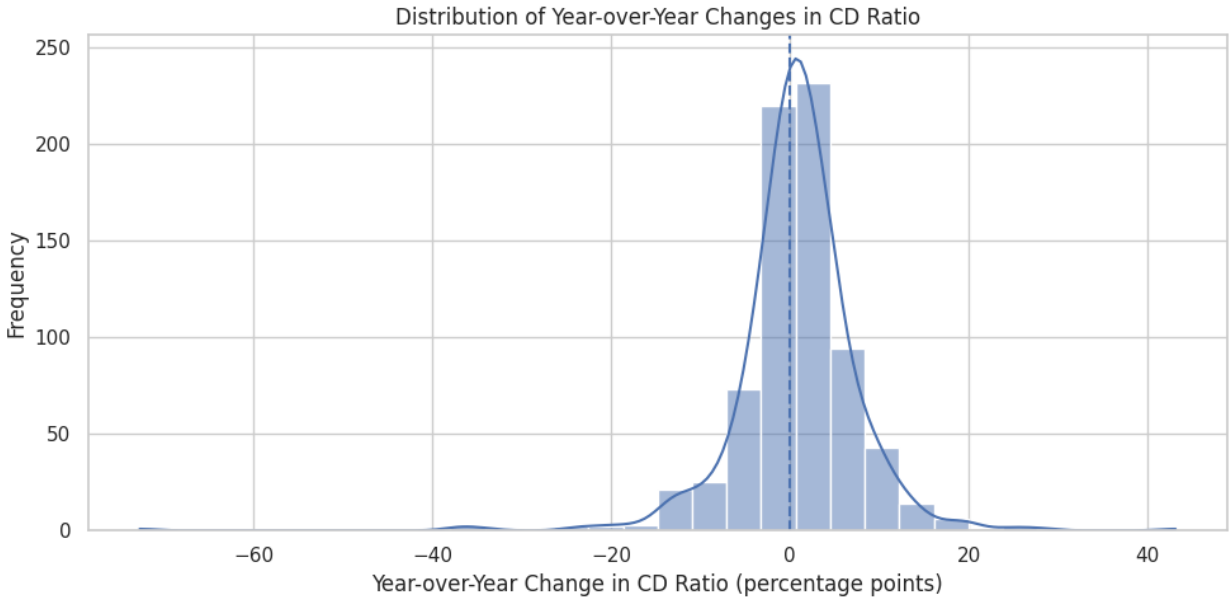


Figure 7. Distribution of year-over-year changes in CD ratio (percentage points).

State	Year	CD Ratio (%)	YoY Change (pp)
Meghalaya	2006	113.9	+43.1
Meghalaya	2007 (decline)	41.2	−72.7
Dadra & Nagar Haveli*	2006	136.7	+25.9
Chandigarh	2009	119.9	+24.1
Nagaland	2012	47.6	+20.1

Table 2. Selected largest single-year increases and the single largest decrease (Meghalaya, 2007) in CD ratio.

Year-over-year changes are centred close to zero and approximately symmetric, but with a small number of extreme swings - mostly in low-population union territories and North-Eastern states where a small change in absolute credit or deposits produces a large percentage swing in the ratio.

4.9 Long-Term Change (2004-2025)

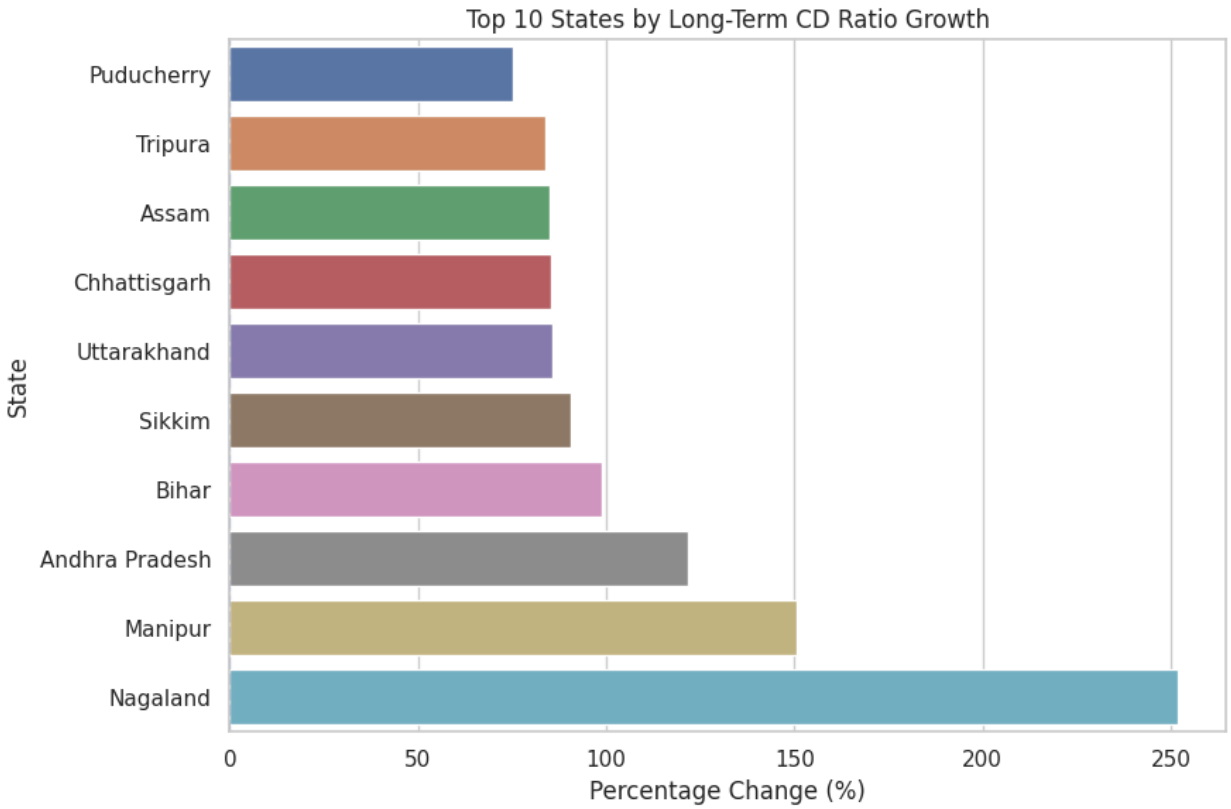


Figure 8. Top 10 states by long-term percentage change in CD ratio, first available year to last available year.

State	First Year → Last	CD Ratio Change	% Change
Nagaland	2004 → 2025	15.2 → 53.5	+252.0%
Manipur	2004 → 2025	31.8 → 79.7	+150.6%
Andhra Pradesh	2004 → 2025	70.9 → 157.2	+121.7%
Bihar	2004 → 2025	26.9 → 53.5	+98.9%
Sikkim	2004 → 2025	23.0 → 43.8	+90.4%

Table 3. Top 5 states by long-term percentage growth in CD ratio.

At the other end of the spectrum, Dadra & Nagar Haveli (−56.1%), Daman & Diu (−40.8%, discontinued after 2019 due to merger) and Chandigarh (−29.9%) recorded the steepest long-term declines, in each case falling back from an unusually high early-2010s peak.

4.10 Regional Comparison

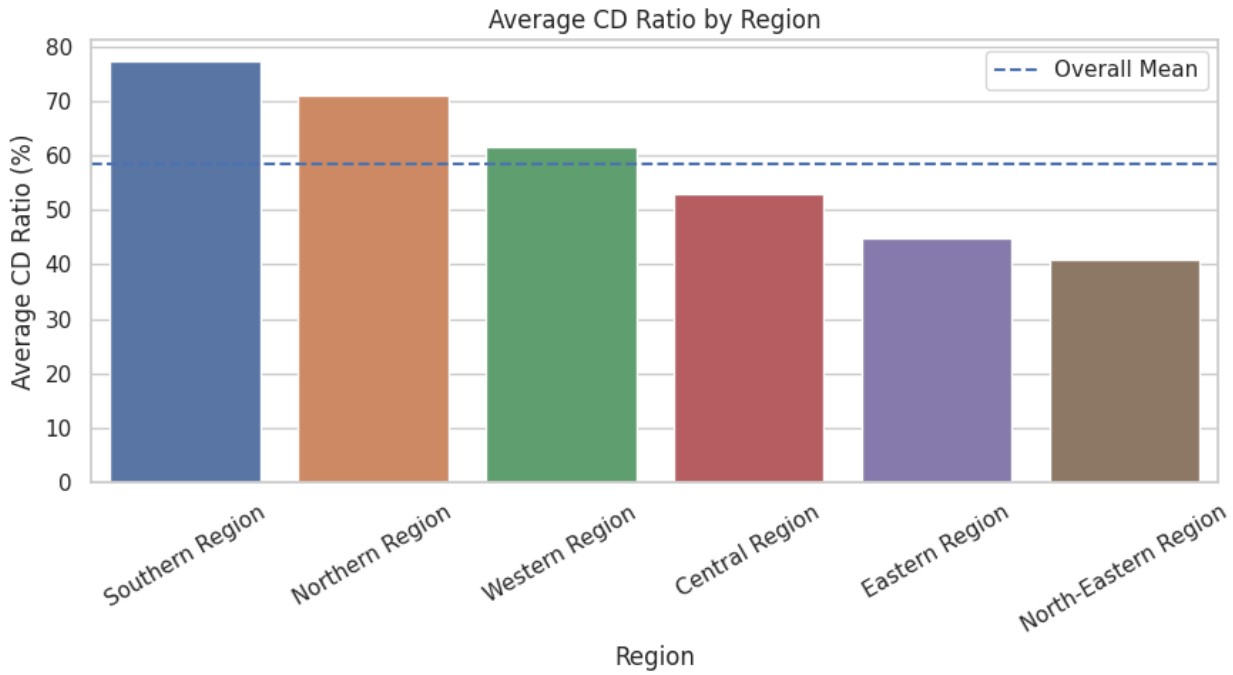


Figure 9. Average CD ratio by region, with the dashed line marking the overall pooled mean (58.53%).

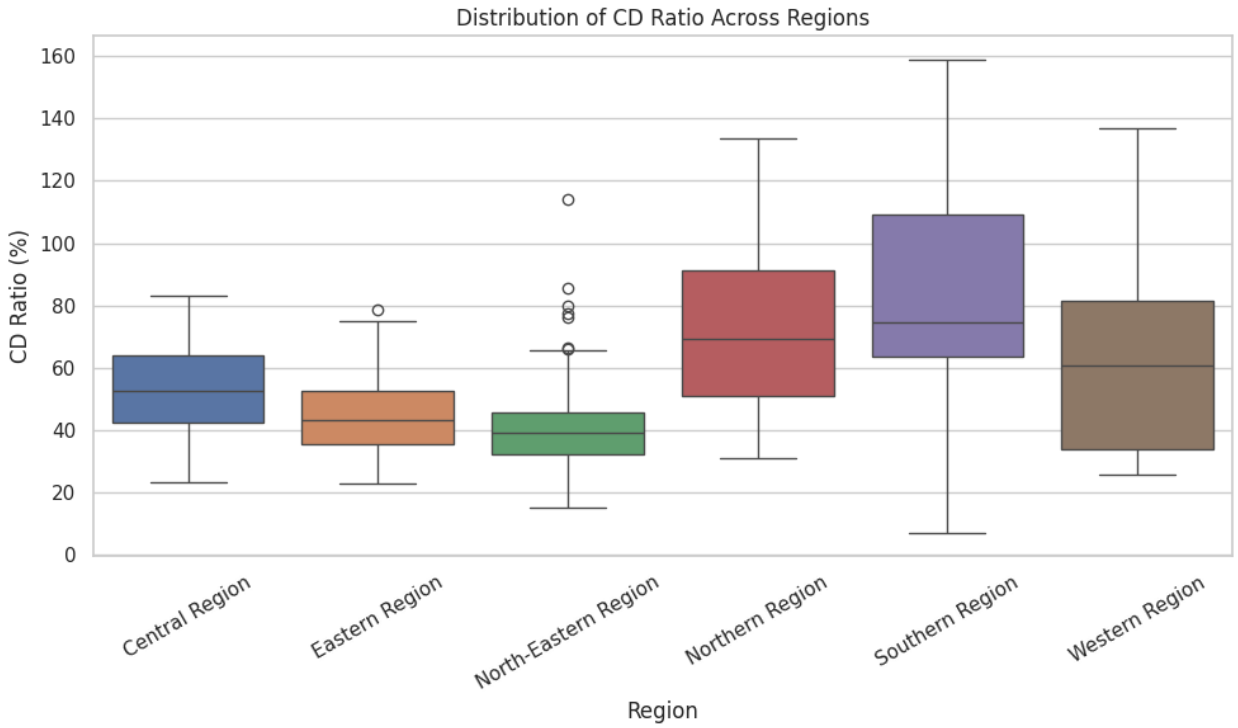


Figure 10. Distribution (box plot) of CD ratio by region.

Region	Mean	Median	Std. Dev.	Min	Max	n
Southern Region	77.25	74.40	36.66	7.1	158.9	143
Northern Region	71.08	69.35	24.44	31.1	133.7	160
Western Region	61.66	60.70	26.64	25.8	136.7	104
Central Region	52.92	52.40	13.91	23.4	83.2	88
Eastern Region	44.76	43.10	11.80	23.0	78.7	132
North-Eastern Region	41.01	39.30	13.22	15.2	113.9	154

Table 4. Regional descriptive statistics for the CD ratio.

4.11 Pre-2015 vs Post-2015 Comparison

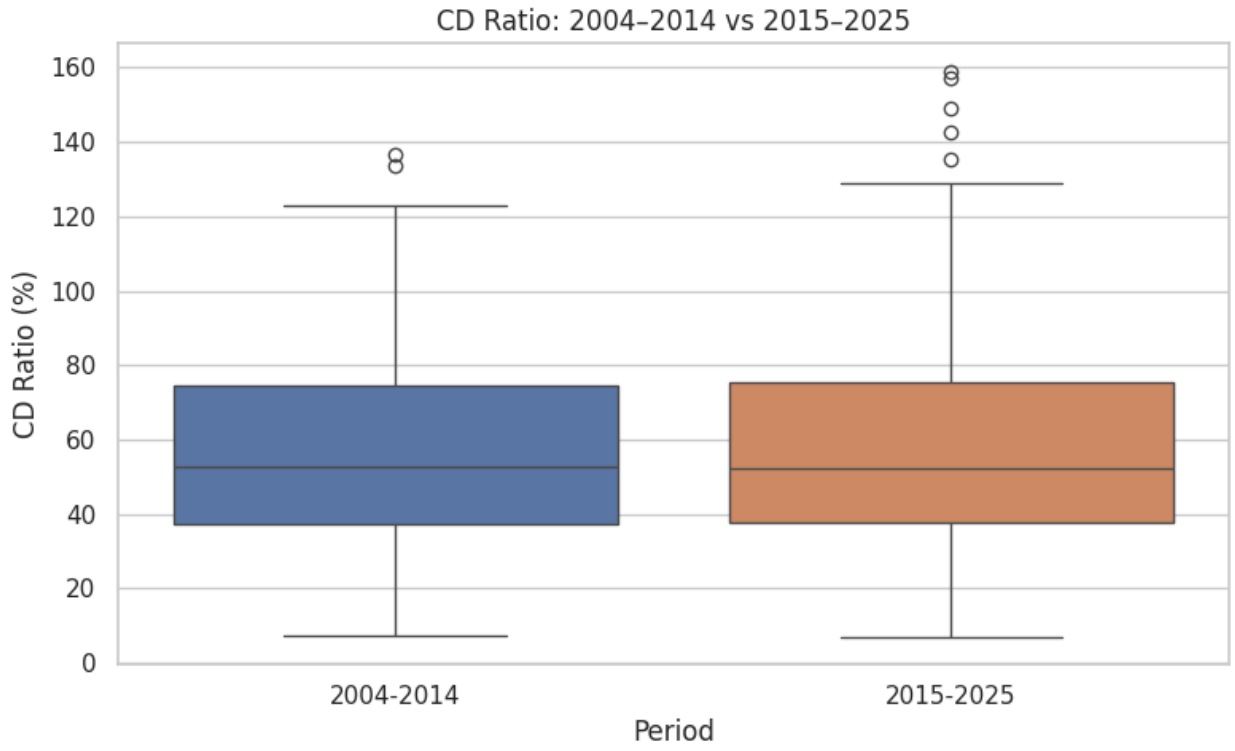


Figure 11. CD ratio distribution, 2004-2014 vs 2015-2025.

The mean CD ratio was 58.10% in 2004-2014 and 58.95% in 2015-2025 - a difference of less than one percentage point at the national/pooled level.

4.12 National Linear Trend

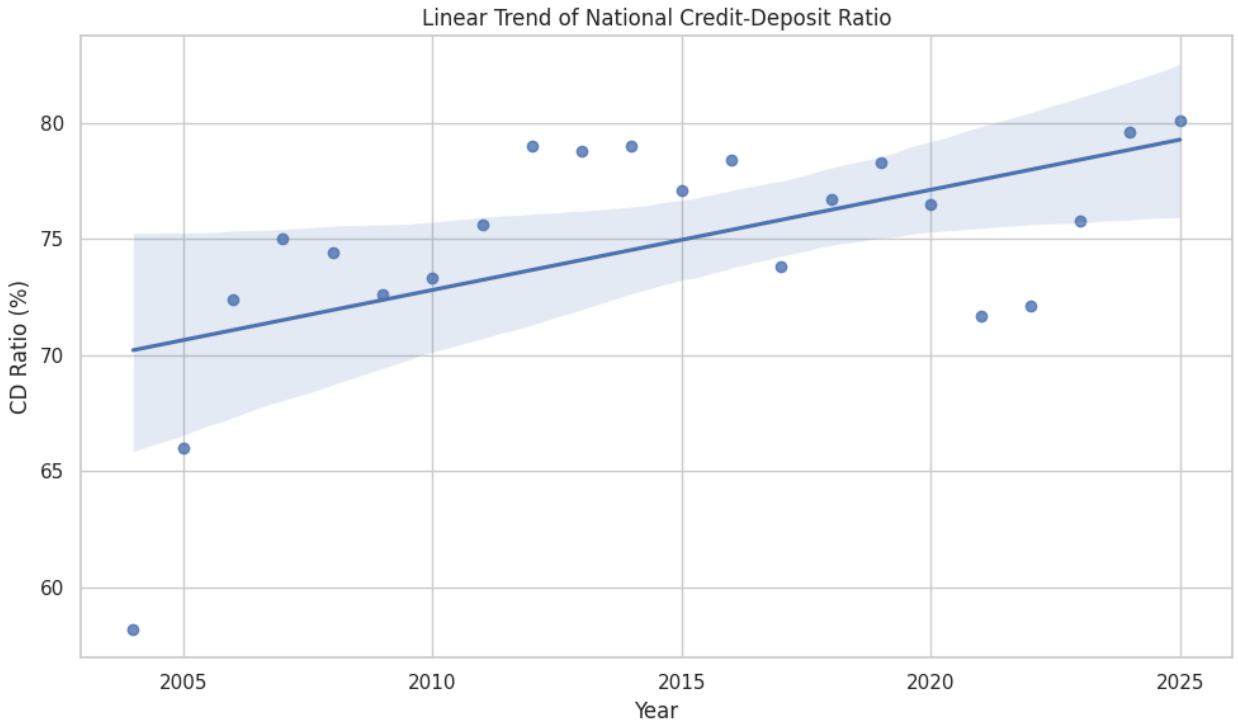


Figure 12. Ordinary least-squares linear trend of the national CD ratio against year (95% CI band shaded).

4.13 Statistical Hypothesis Tests

Statistical Test	Statistic	p-value	Significant (α=0.05)
Paired t-test: pre- vs post-2015 (by state)	t = 0.341	0.735	No
Wilcoxon signed-rank test: pre- vs post-2015	W = 296.0	0.765	No
Pearson correlation: year vs national CD ratio	r = 0.560	0.0067	Yes
Spearman correlation: year vs national CD ratio	ρ = 0.490	0.0207	Yes
National linear trend regression (slope)	slope = 0.431 pp/yr	0.0067	Yes
Kruskal-Wallis: differences across 6 regions	H = 211.71	8.85×10^{-44}	Yes

Table 5. Summary of key statistical test results.

The paired t-test and its non-parametric counterpart (Wilcoxon signed-rank test) both fail to reject the null hypothesis that the mean state-level CD ratio is unchanged between 2004-2014 and 2015-2025 ($p = 0.735$ and $p = 0.765$ respectively). In contrast, both the Pearson and Spearman correlations between year and the national CD ratio are positive and statistically significant ($r = 0.56$, $p = 0.0067$; $\rho = 0.49$, $p = 0.0207$), and the linear trend regression confirms a small but statistically significant upward national trend of about 0.43 percentage points per year ($R^2 = 0.314$). The Kruskal-Wallis test overwhelmingly rejects the null hypothesis of equal distributions across the six regions ($p < 10^{-43}$), confirming that regional disparities identified in Section 4.10 are statistically robust rather than due to chance.

4.14 Post-hoc Regional Comparisons

Because the Kruskal-Wallis test only establishes that at least one region differs from the others, pairwise Mann-Whitney U tests were run for all 15 region pairs. All 15 pairwise comparisons were statistically significant at $\alpha = 0.05$, with the North-Eastern Region vs Northern Region pair showing the strongest difference ($U = 3448.0$, $p = 2.71\times10^{-28}$), and the Central Region vs Western Region pair showing the weakest, though still significant, difference ($U = 3783.0$, $p = 0.039$). This indicates that every region's CD-ratio distribution is statistically distinguishable from every other region's.

4.15 State-wise Trend Regression

An individual linear regression of CD ratio on year was fitted for each state. Of the 37 states/UTs, 23 showed a statistically significant linear trend ($p < 0.05$) - 14 increasing and 9 decreasing.

Rank	State	Slope (pp/yr)	R^2	p-value
1 (steepest rise)	Andhra Pradesh	+3.316	0.859	5.8×10^{-10}
2	Ladakh	+2.043	0.835	0.0109
3	Delhi	+1.888	0.560	6.2×10^{-5}
…	…	…	…	…
Steepest decline	Dadra & Nagar Haveli*	−3.690	0.615	1.6×10^{-5}
2nd steepest decline	Daman & Diu	−1.707	0.724	2.9×10^{-5}
3rd steepest decline	Odisha	−1.314	0.564	5.6×10^{-5}

Table 6. Extremes of the state-wise CD-ratio trend regression (23 states significant at $\alpha = 0.05$; full state-level results are provided in the accompanying analysis CSV files).

4.16 CD-Ratio Benchmark Analysis (>100%)

Region	Years Above 100%	Total Observations	% Above 100%
Southern Region	48	143	33.57%
Northern Region	19	160	11.88%
Western Region	5	104	4.81%
North-Eastern Region	1	154	0.65%
Eastern Region	0	132	0.00%
Central Region	0	88	0.00%

Table 7. Share of state-year observations exceeding the 100% CD-ratio benchmark, by region.

At the state level, Tamil Nadu spent 95.5% of all observed years above the 100% benchmark, followed by Telangana (90.9%, of a shorter reporting window since 2015) and Andhra Pradesh (77.3%). By contrast, several states - including Bihar and the Andaman & Nicobar Islands - never once crossed the 100% benchmark in the full 22-year window, underscoring the persistence of the credit-deployment gap between the highest- and lowest-CD-ratio states.

5. Discussion

5.1 Key Findings

- Persistent regional hierarchy: the Southern Region has led every other region in CD ratio throughout 2004-2025, followed by the Northern Region, while the North-Eastern Region has consistently lagged. This hierarchy is statistically confirmed by the Kruskal-Wallis test and all 15 pairwise Mann-Whitney comparisons, meaning it is not an artefact of a small sample.
- Modest but real national upward drift: although a simple pre-/post-2015 comparison shows no significant shift in mean CD ratio (paired t-test and Wilcoxon test both non-significant), a continuous year-on-year regression detects a small, statistically significant upward trend of roughly 0.43 percentage points per year at the national level ($R^2 = 0.31$). This illustrates why a binary before/after comparison and a continuous trend regression can disagree: the CD ratio has drifted upward gradually and with substantial year-to-year noise, rather than shifting abruptly at a single point in time.
- A concentrated set of states drive the extremes: Andhra Pradesh, Tamil Nadu, Telangana and Chandigarh account for almost all state-years above the 100% benchmark, and also show some of the strongest statistically significant upward trends. Conversely, union territories affected by administrative reorganisation (Dadra & Nagar Haveli, Daman & Diu) show the steepest declines, largely reflecting a return from an artificially high, thinly-based ratio rather than a genuine deterioration in banking-sector health.
- Volatility is concentrated in small/thin-base geographies: states and union territories with a small banking base (Meghalaya, Dadra & Nagar Haveli, the North-Eastern states in general) show the highest year-to-year volatility, because a small number of large loans or deposits can move the ratio disproportionately - an important caveat for anyone using year-over-year CD-ratio changes as a policy signal in these states.
- NoSQL schema flexibility was validated in practice: the single cdRatio collection cleanly accommodated three levels of granularity (national, region, state) without any structural change, and the same document shape survived the historical additions and discontinuations in the underlying reporting units (Telangana's creation in 2014, Ladakh's creation in 2020, the Daman & Diu / Dadra & Nagar Haveli merger in 2020) - exactly the kind of schema churn a fixed relational table would have struggled to absorb without migrations.

5.2 Interpretation and Implications

For policymakers, the persistently low CD ratio in the North-Eastern and Eastern regions signals structurally weaker credit absorption relative to deposits, and is consistent with the same regions' historically higher weight in RBI and NITI Aayog financial-inclusion interventions. For commercial banks and NBFCs, the statistically significant upward trend in a specific set of high-growth states (Andhra Pradesh, Ladakh, Delhi, Manipur, Chhattisgarh) is a more actionable signal for branch-expansion planning than the (statistically insignificant) national pre-/post-2015 shift. For credit-rating agencies, the very high and rising CD ratio in a small number of states (particularly Andhra Pradesh, above 150% by 2025) may warrant closer monitoring of credit concentration risk relative to the local deposit base.

5.3 Limitations

- The CD ratio reflects credit sanctioned/utilised according to place of utilisation, not necessarily the residence of the borrower or lender, so a state with large industrial or infrastructure lending can show an elevated ratio without a correspondingly large local deposit base.
- Administrative reorganisation (state bifurcation, union-territory mergers) introduces structural breaks that a simple linear trend model does not explicitly account for; the very steep slopes observed for Telangana, Ladakh and Dadra & Nagar Haveli should be read in that light.
- This analysis uses a single indicator (CD ratio). The originally scoped $lookup join with a povertyIndicators collection (MPI, unemployment) and the $bucket-based income-tier classification are demonstrated as MongoDB query patterns in Section 3.2, but a full multi-indicator socio-economic analysis was outside the scope of the statistical results reported in Section 4.
- The paired t-test / Wilcoxon comparison of the two periods assumes each state contributes one summary value per period; it does not model within-period year-to-year autocorrelation, which could affect the standard errors.

6. Conclusion

This project demonstrated an end-to-end pipeline for managing and analysing India's state-wise banking data: a flexible MongoDB schema that absorbed three levels of data granularity and 22 years of RBI reporting-unit changes without structural migration; a full set of CRUD, filtering, aggregation ($group, $bucket, $facet), cross-collection join ($lookup) and indexing operations that showed clear efficiency and flexibility advantages over a rigid relational approach; and a rigorous statistical analysis - covering descriptive statistics, year-over-year and long-term change analysis, paired hypothesis tests, correlation and regression, and non-parametric regional comparisons - that uncovered a statistically robust, persistent regional hierarchy in credit deployment (Southern > Northern > Western > Central > Eastern > North-Eastern), a modest but significant national upward trend of about 0.43 percentage points per year, and a small set of states (Andhra Pradesh, Tamil Nadu, Telangana, Chandigarh) that account for almost all instances of credit exceeding deposits at the state level.

These findings, and the underlying MongoDB collection, directly support the real-world use cases outlined in Section 2.2 - from RBI and NITI Aayog policy targeting, to bank branch-expansion strategy, to credit-rating agencies' assessment of sub-national fiscal health - and illustrate concretely why a schema-flexible NoSQL data store, paired with a conventional statistical/ML analysis layer, is well suited to real-world, semi-structured government financial datasets of this kind.

6.1 Future Work

- Complete the $lookup-based integration with the Multidimensional Poverty Index and unemployment-rate collections to formally test the relationship between financial inclusion and poverty reduction across states.
- Extend the schema to additional RBI Handbook tables (bank branches per lakh population, priority-sector credit share, NSDP) to build a fuller multi-indicator state-finance profile.

- Apply time-series or panel-data models (e.g., structural-break detection, fixed-effects panel regression) to more formally account for the administrative reorganisation events identified as a limitation in Section 5.3.
- Build a lightweight dashboard on top of the MongoDB aggregation pipelines developed in Section 3.2 to let policymakers interactively query CD-ratio thresholds, regions and years.

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